

Jackson Burzynski

Curriculum Vitae

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EDUCATION

- 2021 **Ph.D. Physics**, *University of Massachusetts*, Amherst, MA.
 - ADVISOR: Benjamin Brau
- 2018 **M.S. Physics**, *University of Massachusetts*, Amherst, MA.
 - ADVISOR: Benjamin Brau
- 2016 **B.S. Physics & Mathematics**, *Tufts University*, Medford, MA.
 - ADVISOR: Pierre-Hugues Beauchemin
 - *summa cum laude*

PROFESSIONAL EXPERIENCE

- 2021–Present **Postdoctoral Research Fellow**, *Simon Fraser University*, Burnaby, BC.
- 2016–2021 **Graduate Research Assistant**, *University of Massachusetts*, Amherst, MA.

AWARDS AND HONORS

- 2022 **ATLAS Outstanding Achievement Award**, *ATLAS Collaboration*.
Gives recognition to excellent contributions made to the ATLAS collaboration (approx. 3000 scientists). Five to ten awards per year. Awarded for "outstanding contributions to the integration of large-radius tracking into the standard ATLAS reconstruction".
- 2021 **Springer Thesis Prize**, *Springer Publishing*.
For the thesis "A Search for Exotic Higgs Decays or: How I Learned to Stop Worrying and Love Long-lived Particles"
- 2021 **ATLAS Thesis Award**, *ATLAS Collaboration*.
For the thesis "A Search for Exotic Higgs Decays or: How I Learned to Stop Worrying and Love Long-lived Particles"
- 2017 **Summer Research Fellowship**, *University of Massachusetts Amherst*.
- 2016 **Phi Beta Kappa Honor Society**, *Tufts University*.
- 2016 **Amos Emerson Dolbear Scholarship**, *Tufts University*.
- 2014 **Summer Scholars Fellowship**, *Tufts University*.

RESEARCH EXPERIENCE AND PROJECTS

Leadership and Management

- 2024–Present **ATLAS Long-lived and Unconventional Physics Subgroup Convener**.
Coordinate searches for long-lived particles and unconventional signatures (approx. 150 scientists across 20 analysis teams); communicate with group analysts; provide feedback on analysis strategy and implementation; led review of [1, 2].
- 2022–2024 **ATLAS Tracking and Vertexing Subgroup Convener**.
Coordinated all studies of track and vertex reconstruction for Run 3 (approx. 30 scientists); maintained official ATLAS recommendations for tracking efficiency and fake rate uncertainties; supervised ATLAS authorship qualification projects; provided feedback on strategy for publications; led review of [5, 11, 12].

- 2022 **ATLAS Tracking Recommendations Forum Chair.**
Oversaw delivery of operational points and systematic uncertainties for both prompt and displaced track reconstruction for Run 3 (approx. 15 scientists); developed and maintained tracking analysis software tools.
- 2021–Present **ATLAS Tracking Liason to the Exotics Group.**
Disseminate information and give reports on track reconstruction performance and recommendations to the Exotics physics group (approx. 500 scientists).
- 2021–Present **ATLAS Expert Review.**
Expert reviewer and editorial board member for ATLAS publications, including [7]; ensure quality and validity of results; assist in writing public documentation and publications.

Physics Analysis

- 2022–Present **Search for emerging jets in ATLAS.**
Lead analyst in the first search for s -channel production of emerging jets via a Z' mediator using Run 3 data. Designed and implemented a transformer jet tagging algorithm which improved analysis sensitivity by over one order of magnitude.
- 2021–2024 **Search for displaced vertices in the ATLAS Inner Detector.**
Proposer, co-coordinator, and lead analyst for a search for light long-lived particle decays in the ATLAS Inner Detector in Run 2. First ATLAS search results utilizing improved displaced track reconstruction algorithm. Achieved order of magnitude improvement in sensitivity for decays of the Higgs boson to long-lived particles with respect to previous ATLAS results using the same Run 2 dataset. First LHC results for long-lived photophobic axion-like particles. Lead editor of paper [4].
- 2019–2021 **Search for exotic decays of the Higgs boson to long-lived particles in ATLAS.**
Coordinator and lead analyst for the first search for exotic decays of the Higgs boson to pairs of long-lived particles decaying in the ATLAS Inner Detector using Run 2 data. Lead editor of paper [8].
- 2016–2019 **Search for displaced vertices in the ATLAS Inner Detector and Muon Spectrometer.**
Analyst for a search for long-lived particle pairs decaying in the ATLAS Inner Detector and Muon Spectrometer. Lead editor of paper [9].

Track and Vertex Reconstruction

- 2023–Present **Track reconstruction for HL-LHC using graph neural networks.**
Developer of new algorithm for track reconstruction at the HL-LHC using graph neural networks. Assessing algorithmic susceptibility to detector misalignment and optimizing performance for displaced particles.
- 2023–Present **Primary vertex identification using transformers.**
Proposer and lead developer of a transformer network to improve the efficiency of identifying the hard-scatter primary vertex and rejecting pileup interactions.
- 2021–2023 **Large impact parameter track reconstruction.**
Developer of improved algorithm for reconstructing displaced charged particles from long-lived particle decays in the ATLAS Inner Detector. Achieved a 1000% reduction in CPU usage and a 5000% reduction in disk space usage. Led to integration into standard ATLAS reconstruction enabling all ATLAS physics analyses to exploit displaced tracking information. Led first study of performance on data and derived systematic uncertainties on reconstruction efficiency. Coordinator, lead analyst, and lead author of paper [6].

Software and Hardware

- 2023–Present **Tracking Contact to the Analysis Model Group.**
Maintain ATLAS tracking software tools; manage installation of software input data

- 2021-Present **Data formats for unconventional physics.**
 Creator and maintainer of data format to enable approx. 10 long-lived and unconventional physics analyses in Run 2/3. Proposer of a novel strategy for reducing disk usage for analysis data formats. Contributing author for [3].
- 2018-2020 **ATLAS ITk upgrade.**
 Developed and maintained a robotic testing apparatus for performing QA/QC measurements on flex circuits for ITk strip barrel staves.

SELECTED PRESENTATIONS

Invited Speaker at International Conferences and Workshops

- 2024 **Searches for Long Lived Particles at the LHC**, *Dark Interactions 2024*, Vancouver, BC.
- 2024 **Probing the Lifetime Frontier at the LHC and Beyond**, *Physics Poential of Future Colliders*, Vancouver, BC.
- 2024 **Tracking and Vertexing in ATLAS**, *2024 ATLAS-CMS Flavour Tagging Workshop*, Genova, IT.
- 2024 **Multiple track signatures**, *Roadmap of Dark Matter models for Run 3*, Geneva, CH.
- 2022 **What's new for Run 3 for LLPs at ATLAS?**, *Searching for long-lived particles at the LHC and beyond: Twelfth workshop of the LLP Community*, Geneva, CH.
- 2022 **Searches for BSM physics using challenging and long-lived signatures with the ATLAS detector**, *The XXIX International Conference on Supersymmetry and Unification of Fundamental Interactions (SUSY 2022)*, Ioannina, GR.
- 2022 **Improved Track Reconstruction Performance for Long-lived Particles in ATLAS**, *Connecting the Dots (CTD2022)*, Princeton, NJ.
- 2022 **Reduced formats for long lived particles in ATLAS**, *Analysis Ecosystems Workshop*, Orsay, FR.
- 2021 **Search for exotic decays of the Higgs boson to long-lived particles using displaced vertices in the ATLAS inner detector**, *Searching for long-lived particles at the LHC and beyond: Ninth workshop of the LLP community*, Geneva, CH.
- 2021 **Searches for New Long-lived Particles with the ATLAS detector**, *Phenomenology Symposium (PHENO)*, Pittsburg, PA.
- 2019 **Electrical defects in co-cured bus tapes and quality control strategies**, *Forum on Tracking Detector Mechanics*, Ithaca, NY.

Seminars and Colloquia

- 2024 **Long-lived particle searches in ATLAS**, *ATLAS Lecture Series*, Online.
- 2024 **Probing the Lifetime Frontier with the ATLAS Inner Detector**, *University of Washington*, Seattle, WA.
- 2024 **Probing the Lifetime Frontier with the ATLAS Inner Detector**, *University of Oregon*, Eugene, OR.
- 2024 **Probing the Lifetime Frontier with the ATLAS Inner Detector**, *Nikhef*, Amsterdam, NL.
- 2024 **Probing the Lifetime Frontier with the Large Hadron Collider**, *New York University*, New York, NY.

Talks at National Conferences

- 2023 **ATLAS Run 3 Operations and Highlights**, *Canadian Association of Physicists (CAP) Congress*, Fredericton, NB.

Contributed Talks (not as speaker)

- 2024 **Improving Computational Performance of ATLAS GNN Track Reconstruction Pipeline**, *International Conference on Computing in High Energizing and Nuclear Physics (CHEP2024)*, Krakow, PL.
- 2023 **Framework for custom event sample augmentations for ATLAS analysis data**, *International Conference on Computing in High Energizing and Nuclear Physics (CHEP2023)*, Norfolk, VA.

Presentations at Internal ATLAS Meetings

- 2024 **Software and Reconstruction Developments**, *US-ATLAS Summer Workshop*, Seattle, WA.
- 2023 **Hard scatter vertex selection**, *Higgs Workshop 2023*, Tokyo, JP.
- 2023 **Leveraging the FTAG Software Stack Beyond Flavour Tagging**, *2023 ATLAS Flavour Tagging Workshop*, Geneva, CH.
- 2023 **Emerging Jets**, *ATLAS Week*, Vancouver, BC.
- 2022 **Specialized data formats**, *ATLAS Exotics Workshop*, Amsterdam, NL.
- 2022 **Tracking CP report**, *ATLAS Week*, Geneva, CH.
- 2021 **Large Radius Tracking (LRT): Overview and Applications**, *ATLAS Joint Search Workshop*, Geneva, CH.
- 2020 **Long-Lived Particle searches for Moriond**, *ATLAS Physics and Performance Week*, Geneva, CH.

OUTREACH

- 2023-Present **ATLAS-Canada EDI Committee.**
Serve as EDI contact within ATLAS-Canada collaboration; organize and administer demographic surveys; report and track demographic information; educate ATLAS-Canada collaborators on EDI issues; promote an inclusive environment to help collaborators thrive.
- 2020-2021 **HEP Software Foundation Mentor.**
Organized and facilitated two virtual training sessions on computing and data analysis for students and researchers across several experiments in HEP.

ORGANIZATIONAL ACTIVITIES

- 2024-Present **Steering Committee**, *LHC Dark Sector Community*.
- 2024 **Organizational Committee**, *ATLAS Exotics 2024 workshop*, Bolognam, IT.
- 2023 **Session Chair**, *ATLAS HDBS and Exotics 2023 workshop*, Barcelona, ES.

TEACHING EXPERIENCE

As Lecturer

- 2022 **PHYS882 G100: Particle Physics II**, *Simon Fraser University*, Burnaby, BC.

As Teaching Assistant

- 2021 **PHY-440: Intermediate Laboratory A**, *University of Massachusetts*, Amherst, MA.
- 2017 **PHY-182LL: Physics II Lab - Electricity and Magnetism**, *University of Massachusetts*, Amherst, MA.
- 2017 **PHY-181LL: Physics I Lab - Mechanics**, *University of Massachusetts*, Amherst, MA.
- 2016 **PHY-131: Introductory Physics I**, *University of Massachusetts*, Amherst, MA.

STUDENT SUPERVISION

Graduate Students

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| ○ Harshit Choudhary (Simon Fraser) | Search for emerging jets in Run 3 |
| ○ Hamza Hanif (Simon Fraser) | Search for displaced vertices in Inner Detector |
| ○ Paras Pokharel (Simon Fraser) | Transformer tagger for emerging jets |
| ○ Sadaf Kadir (Stanford) | Track reconstruction uncertainties for Run 3 |
| ○ Luke McElhinney (Lancaster) | Secondary vertex reconstruction using transformers |
| ○ Elliot Sampson (Lancaster) | Displaced track reconstruction uncertainties |

Undergraduate Students

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| ○ Randon Hall (Simon Fraser) | Search for emerging jets in Run 3 |
| ○ Michalis Panagiotou (CERN) | Search for displaced vertices in Inner Detector |

PUBLICATIONS

I have more than 450 published articles in refereed journals and an h_{hep} index of 82 as calculated from the publication data-base INSPIRE. The publication policy for ATLAS is that all authors are listed in alphabetical order for each paper. I have included below the publications for which I was either a primary author or made a significant contribution. A full publication list is available on request.

With the ATLAS Collaboration

- [1] ATLAS Collaboration, *Search for neutral long-lived particles that decay into displaced jets in the ATLAS calorimeter in association with leptons or jets using pp collisions at $\sqrt{s} = 13$ TeV*, JHEP **11** (2024) 036, [arXiv:2407.09183](#) [hep-ex].
- [2] ATLAS Collaboration, *Search for light neutral particles decaying promptly into collimated pairs of electrons or muons in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector*, EPJC (submitted) (2024), [arXiv:2407.09168](#) [hep-ex].
- [3] ATLAS Collaboration, *Software and computing for Run 3 of the ATLAS experiment at the LHC*, EPJC (submitted) (2024), [arXiv:2404.06335](#) [hep-ex].
- [4] ATLAS Collaboration, *Search for light long-lived particles in pp collisions at $\sqrt{s} = 13$ TeV using displaced vertices in the ATLAS inner detector*, Phys. Rev. Lett. **133** (2024) 161803, [arXiv:2403.15332](#) [hep-ex].
- [5] ATLAS Collaboration, *Software Performance of the ATLAS Track Reconstruction for LHC Run 3*, Comput. Softw. Big Sci. **8** (2023) 9, [arXiv:2308.09471](#) [hep-ex].
- [6] ATLAS Collaboration, *Performance of the reconstruction of large impact parameter tracks in the inner detector of ATLAS*, EPJC **83** (2023) 1081, [arXiv:2304.12867](#) [hep-ex].
- [7] ATLAS Collaboration, *Search for Heavy Neutral Leptons in Decays of W Bosons Using a Dilepton Displaced Vertex in $\sqrt{s} = 13$ TeV pp Collisions with the ATLAS Detector*, Phys. Rev. Lett. **131** (2023) 061803, [arXiv:2204.11988](#) [hep-ex].
- [8] ATLAS Collaboration, *Search for exotic decays of the Higgs boson into long-lived particles in pp collisions at $\sqrt{s} = 13$ TeV using displaced vertices in the ATLAS inner detector*, JHEP **11** (2021) 229, [arXiv:2107.06092](#) [hep-ex].
- [9] ATLAS Collaboration, *Search for long-lived neutral particles produced in pp collisions at $\sqrt{s} = 13$ TeV decaying into displaced hadronic jets in the ATLAS inner detector and muon spectrometer*, Phys. Rev. D **101** (2020) 052013, [arXiv:1911.12575](#) [hep-ex].

Conference Proceedings and Reports

- [10] P. van Gemmeren et al., *Framework for custom event sample augmentations for ATLAS analysis data*, EPJ Web of Conf. **295** (2024) 03016, <https://doi.org/10.1051/epjconf/202429503016>.
- [11] ATLAS Collaboration, *ATLAS Run 3 charged particle track seed finding performance*, ATL-PHYS-PUB-2023-034, 2023, <https://cds.cern.ch/record/2882156>.
- [12] ATLAS Collaboration, *Primary Vertex identification using deep learning in ATLAS*, ATL-PHYS-PUB-2023-011, 2023, <https://cds.cern.ch/record/2858348>.
- [13] G. A. Stewart et al., *HSF IRIS-HEP Second Analysis Ecosystem Workshop Report*, 2022. <https://zenodo.org/record/7418818>.

- [14] J. Burzynski, *Improved Track Reconstruction Performance for Long-lived Particles in ATLAS*, 2022, <https://zenodo.org/record/8119727>.

Theses

- [15] J. Burzynski, *A Search for Exotic Higgs Decays: Or: How I Learned to Stop Worrying and Love Long-Lived Particles*. Springer, 2023.

Miscellaneous

- [16] S. Malik et al., *Software Training in HEP*, Computing and Software for Big Science **5** (2021), <http://dx.doi.org/10.1007/s41781-021-00069-9>.